

Intelligent Research Paper Analysis

FSM-Based Smart Residential Water Consumption Monitoring System Using Flow Sensors

1. Paper Overview

Title	FSM-Based Smart Residential Water Consumption Monitoring System Using Flow Sensors
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2. Executive Summary

Traditional residential water billing systems in shared apartment complexes split total costs equally, leading to unfair billing and encouraging unnecessary water wastage. This capstone project report presents an intelligent, low-cost smart water monitoring system using a YF-S201 water flow sensor interfaced with a NodeMCU ESP8266 microcontroller. The system converts digital pulse signals generated by water flow into volume measurements in litres using interrupt handling and calibration. Operational flow is systematically controlled using Finite State Machine (FSM) logic across discrete states (Idle, Flow Detection, Processing, Notification, and Reset). Integrated with Wi-Fi and Telegram Bot API, the system enables users to monitor real-time, hourly, daily, and monthly water consumption remotely via smartphone commands.

3. Research Problem

In shared residential buildings and apartments, water charges are often billed equally as part of general maintenance fees without accounting for individual household usage. This results in inaccurate billing, lack of transparency, and unfair cost distribution. Furthermore, traditional water meters require manual reading, which is labor-intensive, error-prone, and lacks real-time visibility, automated leak alerts, or remote accessibility. Existing commercial smart meters are too expensive for small-scale residential adoption.

4. Objectives

- Develop a real-time residential water consumption monitoring system using water flow sensors.
- Interface the YF-S201 flow sensor with a NodeMCU ESP8266 microcontroller for pulse detection and processing.
- Implement Finite State Machine (FSM)-based control logic for structured state management.
- Convert sensor pulse signals into water usage values in litres using an adjusted calibration factor.
- Establish wireless IoT communication via IEEE 802.11 Wi-Fi and Telegram Bot API for remote smartphone monitoring.
- Provide Telegram command functionality (/start, /live, /hour, /day, /month, /reset, /status, /help) to retrieve consumption statistics and reset values.
- Build a low-cost, user-friendly prototype model using a water tank setup suitable for residential utility applications.

5. Methodology

The study uses a hardware-software co-design approach. A physical prototype was constructed using a plastic container as a water tank connected to a YF-S201 water flow sensor attached to the outlet line. The sensor's signal pin is wired to digital pin D5 of the NodeMCU ESP8266 microcontroller. As water flows, the sensor's internal rotor rotates and generates electrical pulse signals. The microcontroller counts these pulses using interrupt-driven programming (RISING mode on pin D5) to minimize CPU overhead. A calibration factor converts raw pulse counts into water volume in litres. The operational architecture is governed by FSM logic divided into five states: Idle State, Flow Detection State, Processing State, Notification State, and Reset State. Using built-in Wi-Fi and the Telegram Bot API, the NodeMCU handles command requests from users and transmits real-time consumption data.

6. Dataset

Not clearly mentioned in the paper. The project uses live hardware pulse data generated during physical testing of the water tank prototype setup rather than a pre-existing public dataset.

7. Algorithms & Models

- Finite State Machine (FSM) Control Logic (Idle, Flow Detection, Processing, Notification, Reset states)
- Interrupt-Driven Pulse Counting Algorithm
- Pulse-to-Litre Calibration Algorithm
- Telegram Bot API Communication Protocol over IEEE 802.11 Wi-Fi

8. Key Findings

- Interrupt-driven pulse counting accurately captures rotor rotation signals from flow sensors without overloading the microcontroller CPU.
- Changing the signal pin connection from pin D2 to pin D5 and testing interrupt modes (RISING) significantly improved pulse detection reliability.
- FSM methodology provided organized, predictable, and error-free execution across different operational states.
- Telegram Bot API integration offered an accessible and practical solution for low-cost, real-time IoT remote monitoring on smartphones.
- Continuous feedback on consumption patterns increases user awareness and promotes responsible water usage.

9. Results

The prototype successfully detected water flow, counted pulses, and calculated consumption in litres in real time. Telegram bot commands responded accurately to user queries. Sample test outputs achieved values such as live flow rates of 5.55 L/min, 12.17 L/min, and 15.22 L/min; daily consumption values of 0.70 Liters, 0.81 Liters, and 2.5 Liters; last hour usage of 1.2 Liters; and monthly usage of 15 Liters. The /reset command effectively zeroed current counts while temporarily retaining previous records.

10. Limitations

- Data is stored temporarily in microcontroller RAM and is lost upon power failure or system restart.
- The current prototype only supports single-flow line monitoring and does not feature multi-household tracking.
- Lacks automated leak detection algorithms and automated billing generation in its baseline prototype.
- Requires continuous active Wi-Fi and internet connectivity for Telegram message delivery.

11. Research Gap

The absence of low-cost, open-source residential water monitoring solutions that integrate FSM-based digital design, interrupt-driven pulse processing, and instant smartphone messaging without relying on expensive industrial metering systems.

12. Future Scope

- Integration with cloud databases (Firebase, Google Sheets, ThingSpeak) for persistent historical data storage and analytics.
- Expansion to multi-household/multi-apartment tracking using centralized microcontroller networks.
- Development of a dedicated mobile app featuring graphical dashboards and billing logs.
- Implementation of automated leakage detection and emergency user alerts.
- Automated billing system integration based on consumption tiers and local water tariff rates.
- Integration of Artificial Intelligence and Machine Learning models to forecast usage patterns and detect anomalous consumption.

13. Key Concepts

- Finite State Machine (FSM)
- Interrupt Handling / Interrupt-Driven Processing
- Water Flow Sensor (YF-S201) & Hall-Effect Pulse Generation
- NodeMCU ESP8266 / Embedded System Design
- Internet of Things (IoT) Communication
- Telegram Bot API
- Sensor Calibration

14. Overall Evaluation

The report presents a well-executed capstone project that successfully demonstrates the application of digital system design and IoT principles to address unfair water billing and wastage in shared residential settings. The solution is practical, low-cost, and technically sound, offering strong educational value and clear potential for real-world scaling into smart apartment utility management systems.